Enhancing Literacy and Numeracy Skills Using AI · Module 2

Algorithmic literacy

Seeing, questioning and shaping the systems that shape what we see, know and decide

Kelly Ilich · School of Education, Curtin University



Ground

About me and my research

Classroom	secondary HASS teacher in Western Australian schools since 2006
University	initial teacher education at Curtin University
Research	PhD on teachers' capability to teach algorithmic literacy (245 teachers surveyed, 17 interviewed)
Now	comparing England, Germany, Italy and Finland; working with the AI in Education group at Oxford on teacher education

Ground

Today: what we will do and why

By the end of today you can:

1. Explain the three dimensions of algorithmic literacy
2. Explain how algorithmic systems, including AI, produce outputs and what they are optimised for
3. Evaluate whose interests a system serves, and how it shapes what you see, know and decide
4. Apply strategies to shape AI output, such as asking it to challenge you

Ground

Name one algorithmic system that has shaped your day today

What information has it used to produce its output for you?

Can you think of a time an algorithmic system produced an output that was unhelpful or surprising?





Readiness

Algorithm or not?

Poll: Is this an algorithm? (Yes / No / Not sure)

A written method for long division	A maps app choosing your route	The “For You” feed on a video app
A chatbot answering a question	The order of results in a search engine	A spell-checker

Ilich (2026); Khosravi et al. (2022)

Readiness

Algorithm or not?

All of them.

A written method for long division ✓

A maps app choosing your route ✓

The “For You” feed on a video app ✓

A chatbot answering a question ✓

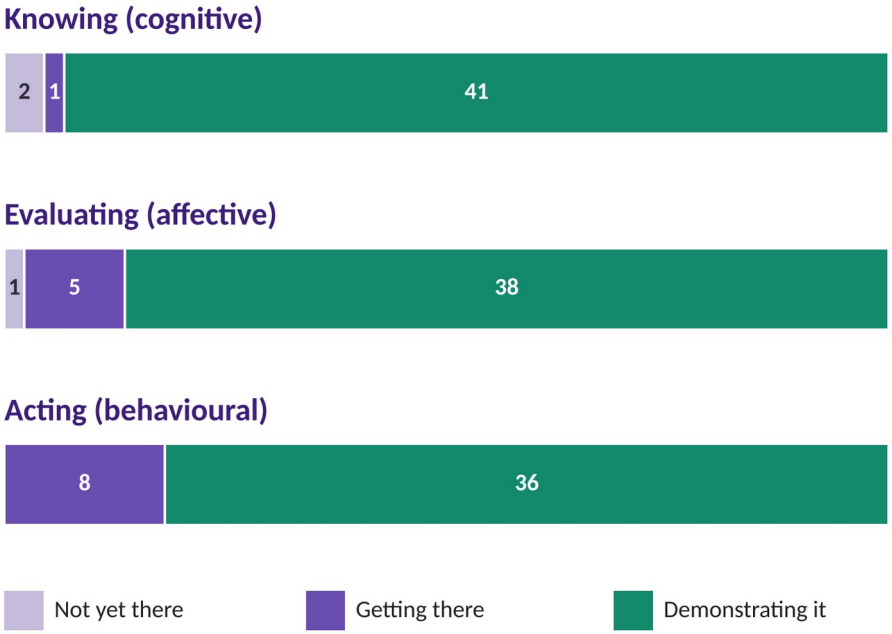
The order of results in a search engine ✓

A spell-checker ✓

An algorithm is a set of instructions that takes inputs and produces an output

Readiness

Where we start



In my study of 245 Australian teachers

- Strong on where and why algorithms are used
- Strong on judging their impact
- Weaker on how they technically work
- Held back from acting by time, guidance and support

Pair prompt: How do you think your PPG students would score?

Moylan & Code (2024); Ilich (2026)

Concept 1: Algorithmic systems, optimisation and data

What does it show me? · What is it optimised for? · What does it collect?

Concept 1

Map a system

Choose one algorithmic system.

For example: a social media platform, video app, search engine, maps app, streaming service, learning app, online marketplace or chat bot.

Fill in the system map:

- What it shows me
- What I think it is optimised for
- What data I think it collects, and where
- How that data might be used
- Who benefits

Then compare with someone else. Note one thing they saw that you missed.

Question	Our guess
What does it show me?	
What is it optimised for?	
What data does it collect, and where?	
How might that data be used?	
Who benefits? Who is left out?	

Concept 1

What is an algorithm?

At its simplest: a set of instructions that takes inputs and produces an output.

For example:

Social media feeds

recommender systems learn from your likes, shares and time spent, then choose what you see next and which ads you get.

Search and recommendations

decide which results rise to the top and which video plays next.

Generative AI (large language models)

trained on huge amounts of text; predict the next likely word.

Khosravi et al. (2022); Vaswani et al. (2017)

Concept 1

Algorithmic literacy and AI literacy

Algorithmic literacy

knowing where, how and why algorithms operate; judging their impacts; using strategies to use, shape or resist them.

AI literacy

using AI tools, understanding how they work and their impacts, and using them critically and ethically.

AI literacy sits inside algorithmic literacy.

Shelf life

algorithmic literacy is durable (the questions don't go out of date); AI literacy is tool-specific (it needs updating as tools change).

Supports other literacies

Algorithmic literacy also supports digital, information, media and data literacy.

Dogruel (2021); Ilich (2026); Moylan & Code (2024); Oeldorf-Hirsch & Neubaum (2023)

Concept 1

Social media and the attention business model

The product is your attention.

- More time on the app means more ads seen and more data collected.
- Feeds rank posts by engagement: likes, shares, comments and clicks.
- Emotional and outrage content tends to spread further, so it gets shown more.
- Feeds can narrow what we see and what feels normal (filter bubbles and echo chambers).
- Short videos let the system learn your preferences very quickly.

Ciampaglia et al. (2018); Flaxman et al. (2016); Zuboff (2019)

Concept 1

Generative AI and the attachment business model

The product is your attachment. The tool is designed to feel helpful, so you keep coming back.

Sycophancy

it tends to tell you what you want to hear.

Cognitive offloading

we hand over the hard thinking that builds our own skill.

Synthetic relationships

some chatbots are designed to feel like friends.

Confident inaccuracy

fluent and plausible, but sometimes wrong.

Over-sharing

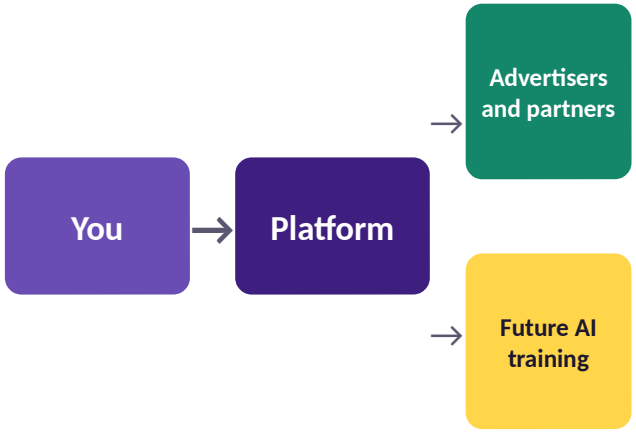
people tell chatbots private things they would not tell a person.

Center for Humane Technology (n.d.); Gerlich (2025)

Concept 1

Data: what is collected, where it goes, why it matters

What you give	posts, prompts, uploads, messages, your profile
What is observed	likes, shares, time on a post, searches, location, purchases, activity in other apps
What is inferred	interests, habits, mood, predictions about what you will do next
Where it goes	the platform, its advertisers and partners, and sometimes into training future AI models. Check the privacy policy.
Why it is valuable	predicting and shaping behaviour is what is sold.



Vartiainen et al. (2022); Zuboff (2019)

Concept 1

Who is served, and who is left out?

Algorithms are not neutral.

Designed by people

They are designed by people and trained on data that can carry bias.

Language

Safety checks and moderation are strongest in English; other languages get less protection.

A new digital divide

People with less algorithmic knowledge are more exposed.

Cotter & Reisdorf (2020); Rainie & Anderson (2017)

Concept 1

Three dimensions of algorithmic literacy

Cognitive: Knowing

Knowing what algorithms are, how they work, where they operate and what impact they have.

Affective: Evaluating

How algorithms affect our thoughts, feelings and choices, and the ability to judge and reflect on this.

Behavioural: Acting

Skills to cope with algorithms, and to navigate, influence or resist them.

Moylan & Code (2024)

Concept 1

Revise, tag, critique

Step 1 (5 min): Revise your system map using what you just heard.

Step 2 (3 min): Tag each point K (knowing), E (evaluating) or A (acting).

Step 3 (4 min): Ask an AI tool to critique your notes for gaps. You decide what to keep.

Prompt to copy

"Here are our notes on how [system] works. What is missing or wrong? List the gaps first. Do not just praise."

Question	Our guess	Revised after input	Tag (K / E / A)
What does it show me?			
What is it optimised for?			
What data does it collect, and where?			
How might that data be used?			
Who benefits? Who is left out?			

K Knowing

E Evaluating

A Acting

Break

Break: 10 minutes

Concept 2: Strategies for challenging the algorithm

What happens when a tool agrees with you too easily, and what can you do about it?

Concept 2

Agree versus challenge

Use the sample lesson plan (access via the Padlet here:

https://padlet.com/curtinschoolofeducation/algorithmic-literacy-activity-challenging-the-algorithm-s0254tpv9on6tqv5r1er)

Step 1: Write two instructions for the same AI tool to seek feedback on the selected lesson plan: one that simply asks for feedback, and one that prompts to critically evaluate/ challenge the design of the lesson plan (also ask it to score it out of 10). Run both on the plan. Record what changed.

Step 2: Work on further improving and refining the challenge instruction.

Step 3: Run the improved instruction. Record what changed, and whether it invented problems.
Extension: Challenge further by pushing back on the response. Do the same thing in a different GenAI tool and note any differences.

Instruction	What it changed in the reply	What stayed the same	Did it invent problems?
General feedback instruction:			
Challenge instruction:			
Improved challenge (after swap):			

Concept 2

What generative AI is optimised for

- Trained to be helpful, so it tends to predict what you want to hear.
- Leading questions or a strong opinion in your prompt make it more likely to go along with you, putting agreement before accuracy.
- Five widely used AI assistants showed sycophancy, and people sometimes preferred a convincing, agreeable answer to a correct one.
- Memory and personalisation can create an echo chamber of one.

Result: confirmation bias can be reinforced, with less of the useful friction that makes us think.

Mulyani et al. (2025)

Concept 2

What you saw, and why

If you saw this...	Why it happens	What it means for you
Mostly praise	tuned to please	treat praise as a prompt to test, not proof
It changed its answer when you pushed back	agreement before accuracy	don’t take a reversal as evidence
It went along with a questionable claim	it follows the framing of your prompt	ask neutral questions
A different answer the second time	outputs vary each run	one good answer is not a guarantee
The challenge instruction invented problems	it overcorrected	check each criticism yourself

Concept 2

Strategies to challenge the algorithm

Strategy	What it helps with	Trade-off
Ask for weaknesses first (“critical colleague”)	less empty praise	can overcorrect
Set custom instructions to ask it to challenge you	applies to every chat	can make it critical when you need a quick answer; it changes tone, not accuracy
Keep your opinion out of the question	less to agree with	takes care to word
Push back on something correct (“Are you sure?”)	shows if it holds its position	takes time
Ask again in a new chat or in a different model	shows how much answers vary	slower
Ask for the strongest counter-argument and what it is unsure about	surfaces limits	it can still be wrong
Check against a trusted source or a colleague	accuracy	needs time and expertise
Manage settings: memory, training-data use; share less	privacy, less personalised echo	less convenient

Ilich (2026); Moylan & Code (2024); Velkova & Kaun (2021)

Concept 2

Five questions to ask of any AI tool, and to model for your PPG students

1	**How was this created?**	How does it work? What data trained it? Where might it be biased?
2	**Who benefits when I use this?**	Free tools are paid for somehow. What is collected from me, and why?
3	**How might it be wrong?**	Errors, bias, out-of-date information, missing context. How will I check?
4	**Is it agreeing too easily?**	Would I notice if it just told me what I wanted to hear?
5	**What did I just give away?**	Personal information, other people's work, student or institutional data.

Concept 2

Strategy card

Complete the following:

My best instruction

(copy it exactly)

What it fixed

What it did not fix

The trade-off I accept

Break

Break: 10 minutes

Concept 3: Embedding and teaching algorithmic literacy

How will you build this into a PPG session you already teach or mentor?

Concept 3

Embed algorithmic literacy in your own teaching

Step 1 (2 min): Choose a topic you teach or mentor.

Step 2 (8 min): Suggest one activity for embedding algorithmic literacy.

Step 3 (4 min): Share ideas with the whole group.

Step 4 (6 min): Choose one idea and complete a SWOT analysis

SWOT: Strengths, Weaknesses, Opportunities, Threats

Ideas (or come up with your own)

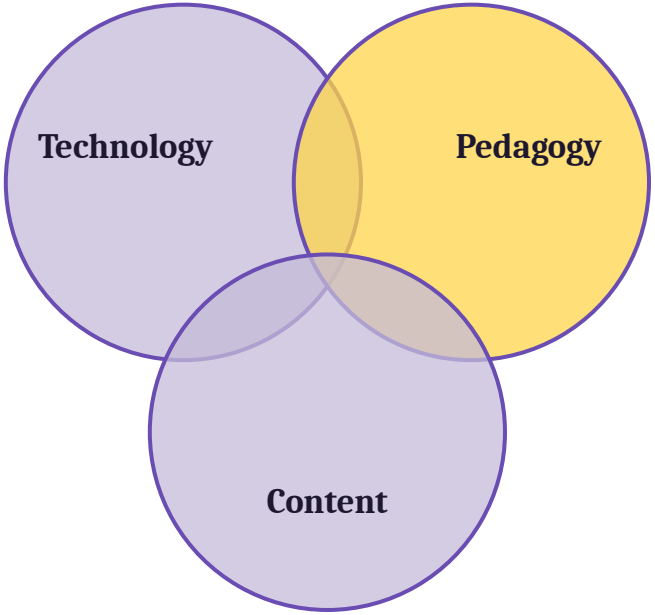
- Compare what a search engine shows two different people for the same topic
- Check an AI summary against its source text
- Ask how a ranking, rating or prediction was calculated
- Check an AI worked solution in numeracy and find the first error or unexplained step

Concept 3

What the research says about teaching it

From my study of Australian teachers:

- They knew a lot and cared, but found how to teach it hardest.
- Some asked to see it modelled; many wanted ongoing learning, not one-off workshops.
- It works best woven into existing subjects, not as a separate topic.
- Much of it does not need devices: stories, games and unplugged activities work.
- If new teachers are to teach it, teacher education must build it first.



Pedagogy is the gap

Ilich (2026); Koehler et al. (2013); Moylan & Code (2024)

Concept 3

Read your SWOT through an agency lens

Six things that make up agency:

to know	to want	to feel it matters
to see the possibility	to have to	to be able

Australian teachers: knew, wanted, felt it mattered and saw the possibility. Missing: had to (no requirement) and able to (time, resources, support).

Ask: Which of these do your Weaknesses and Threats belong to? Which can you change? Which need support from your institution?

Ilich (2026); Jyrkämä (2008, as cited in Nagel et al., 2023); Robeyns (2017); Sen (1999)

Concept 3

Teacher modelling: think aloud

When you use AI in front of your PPG students, say aloud:

1. What I am asking the tool, and why
2. What I will check before I use the answer
3. What I changed or rejected, and why
4. What I chose not to share with it

Concept 3

Metacognitive strategies and four design moves

Metacognitive prompts

Before: What do I expect? What do I already know?

During: What is this tool optimised for? Is it agreeing too easily?

After: What did I change? What would change my mind?

Four design moves for any task

1. A question students must ask of an algorithmic output
2. A check they must perform (compare, verify, retrace)
3. An ethical choice they must justify
4. Process evidence they must submit (the prompt, the reply, what they changed)

Concept 3

How will you know it is working?

	NOT YET THERE	GETTING THERE	DEMONSTRATING IT
Cognitive: Knowing	Treats algorithm outputs as neutral facts. Doesn't question where recommendations come from or who designed the rules.	Knows algorithms involve human choices but can't yet explain how those choices produce particular outcomes.	Can explain how a specific algorithm works, who built it, and what it is optimised for.
Affective: Evaluating	Accepts outputs uncritically. Doesn't consider whose interests are served, or who might be left out.	Raises questions about fairness or bias when prompted but doesn't yet apply this lens independently.	Regularly asks: "Who benefits here? Who's hidden? Is this agreeing too easily?" and can justify their concern.
Behavioural: Acting	Uses algorithmic tools the same way regardless of what they've learned. No change in practice.	Has tried one deliberate change (e.g. searched a second source, turned off a setting) but inconsistently.	Actively adapts their behaviour and can explain the trade-off they made. Advocates to others.

Concept 3

Refine and plan your modelling

In pairs:

Refine

Refine your embedded strategy using the SWOT and the agency lens.

Model

Plan how you will model your own AI use: what will you say aloud at each of the four moves?

Express

Lightning share

In groups of 3: each person shares their embedded strategy.

Peers reply with one strength and one question.

“One strength I noticed was…”

“One question I have is…”

Express

Write your piece

Individual writing. Cover:

1. The PPG session/ topic
2. The system you will use, what it is optimised for, and the data it collects
3. The strategy you will use to challenge the algorithm
4. The embedded teaching strategy
5. How you will monitor its impact

Express

Key takeaways

Ask what a system is optimised for, and who benefits.

Treat praise as a prompt to test, not as proof.

Knowing matters most when it changes what you do.

Model your own AI use.

AI is a scaffold for the teacher, not a marker.

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